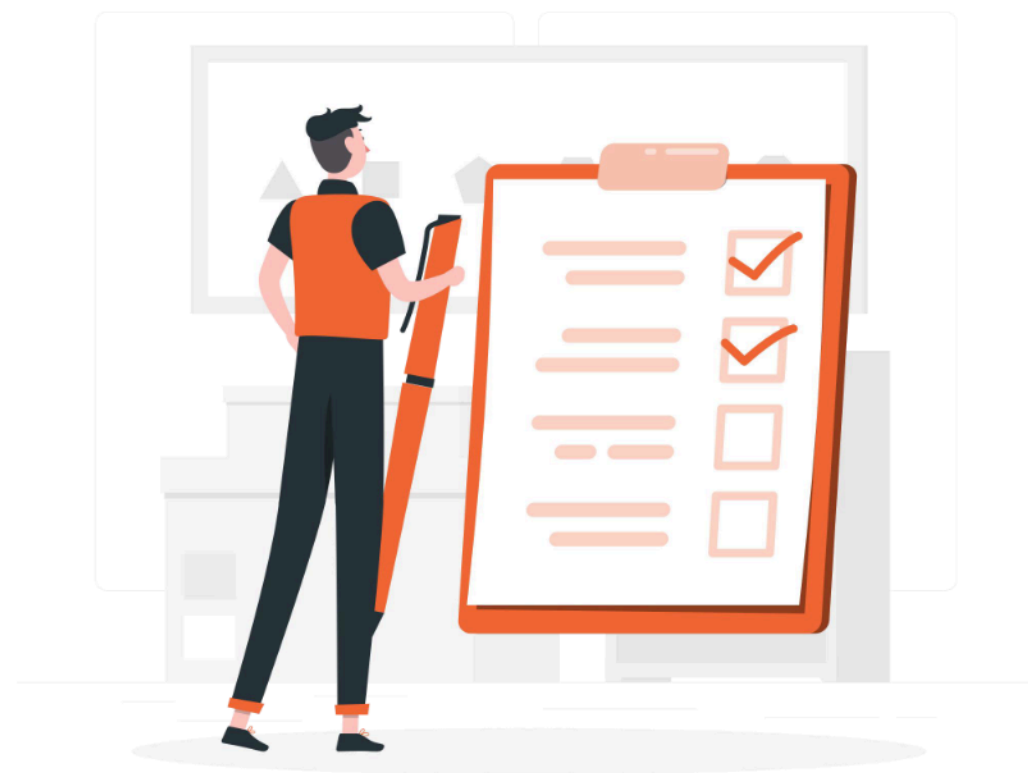




LLM API Integration & Architecture

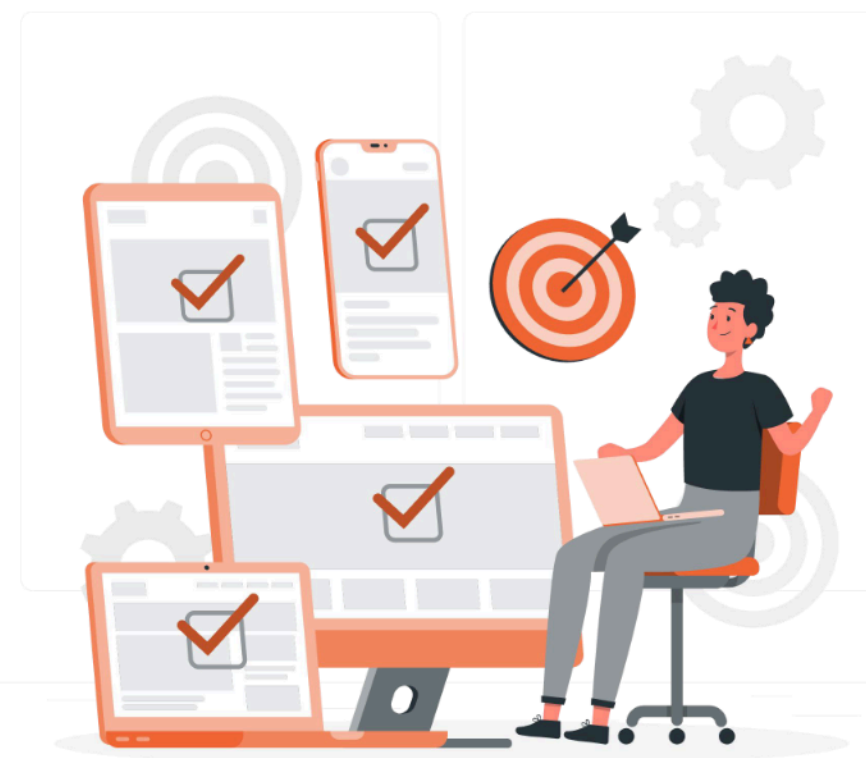
Lesson

1. Calling Azure OpenAI API
2. Embedding generation
3. Token counting & cost estimation
4. Processing large context inputs
5. Vector database workflow



Learning Outcome

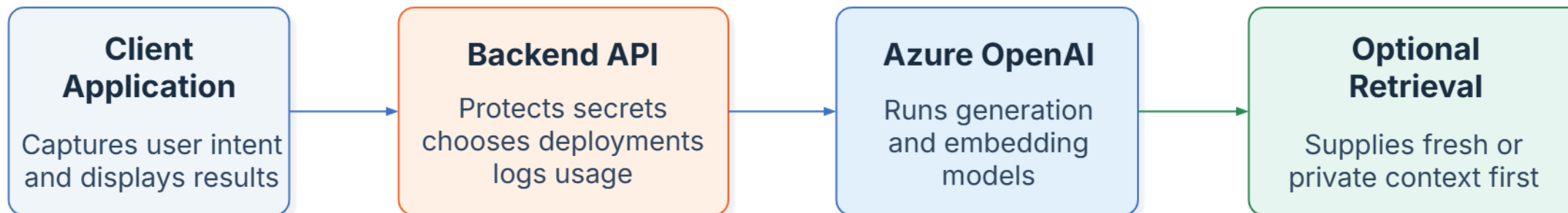
1. Use Azure OpenAI API for generation and embeddings.
2. Manage tokens, costs, and large-context inputs efficiently.
3. Apply vector database workflows for retrieval and AI applications.



From Prompt to Product

Production Changes the Problem

A frontend can collect intent, but the backend is where routing, secrets, logging, fallback logic, and response shaping belong.



Control

The backend decides deployment, instructions, and exposed response fields.

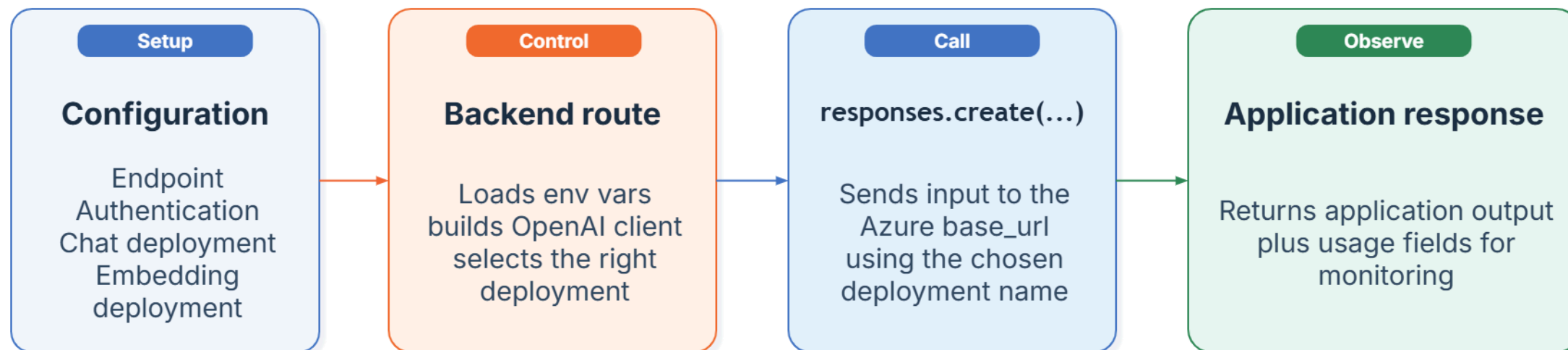
Observability

Routes can record tokens, latency, failure patterns, and request volume.

Extensibility

Retrieval, caching, and orchestration can be added without rewriting the client.

Azure Integration



Key operational reminder

In Azure, the app usually calls the deployment name. That is the application-facing identifier the backend needs to know.

Authentication choices

API keys are common in demos. Microsoft Entra ID matters when teams want keyless, centrally managed authentication.

Why the backend owns it

The client should not know secrets or deployment routing. Keeping those decisions server-side preserves flexibility.

Calling Azure OpenAI API



Azure OpenAI: Overview

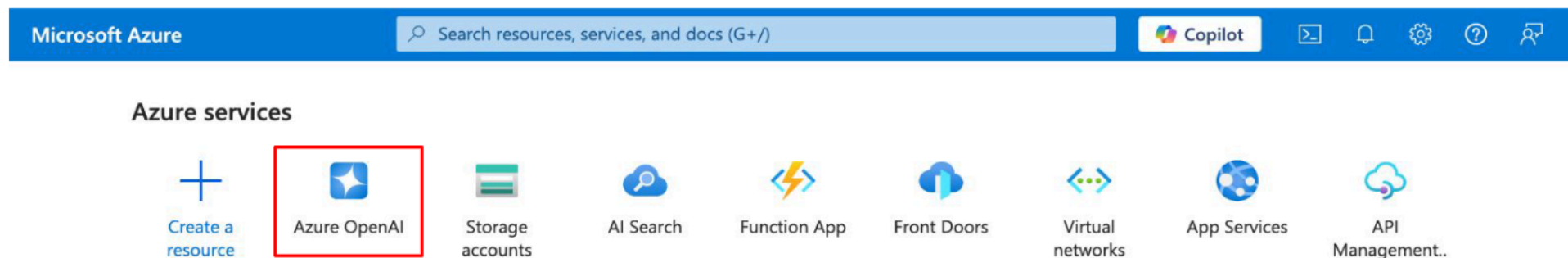
Azure OpenAI is a cloud service from Microsoft that enables the use of advanced AI models such as those from OpenAI, hosted securely on the Microsoft Azure platform.



Feature	OpenAI	Azure OpenAI
Infrastructure	Managed by OpenAI independently	Managed by Microsoft (Azure)
Data Privacy	Customer data used for training	No customer data used for training
Compliance	OpenAI Enterprise standard	Full Azure compliance (HIPAA, SOC2)
Support/SLA	Platform support	Enterprise-grade SLAs
Security	Standard API security	VNETs, RBAC, Private Links

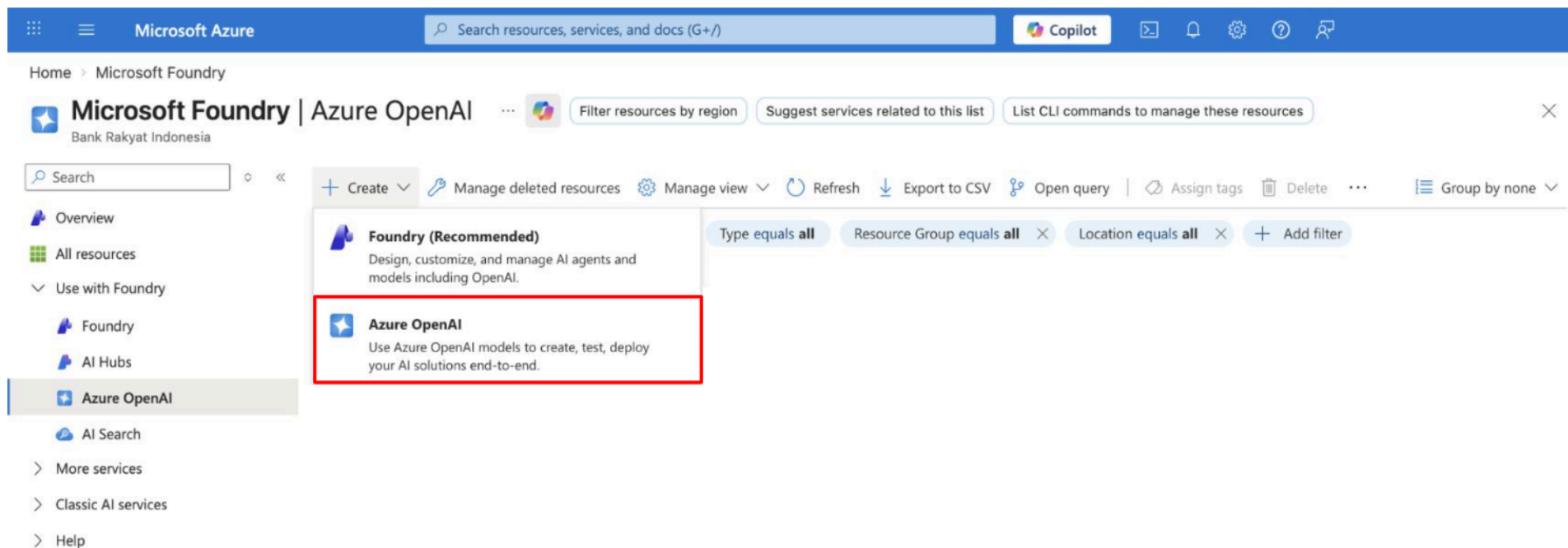
Azure OpenAI: Create Instance

- Visit URL <https://portal.azure.com/>
- Login with your email and password
- It will open the azure portal
- Click **Azure OpenAI**



Azure OpenAI: Create Instance

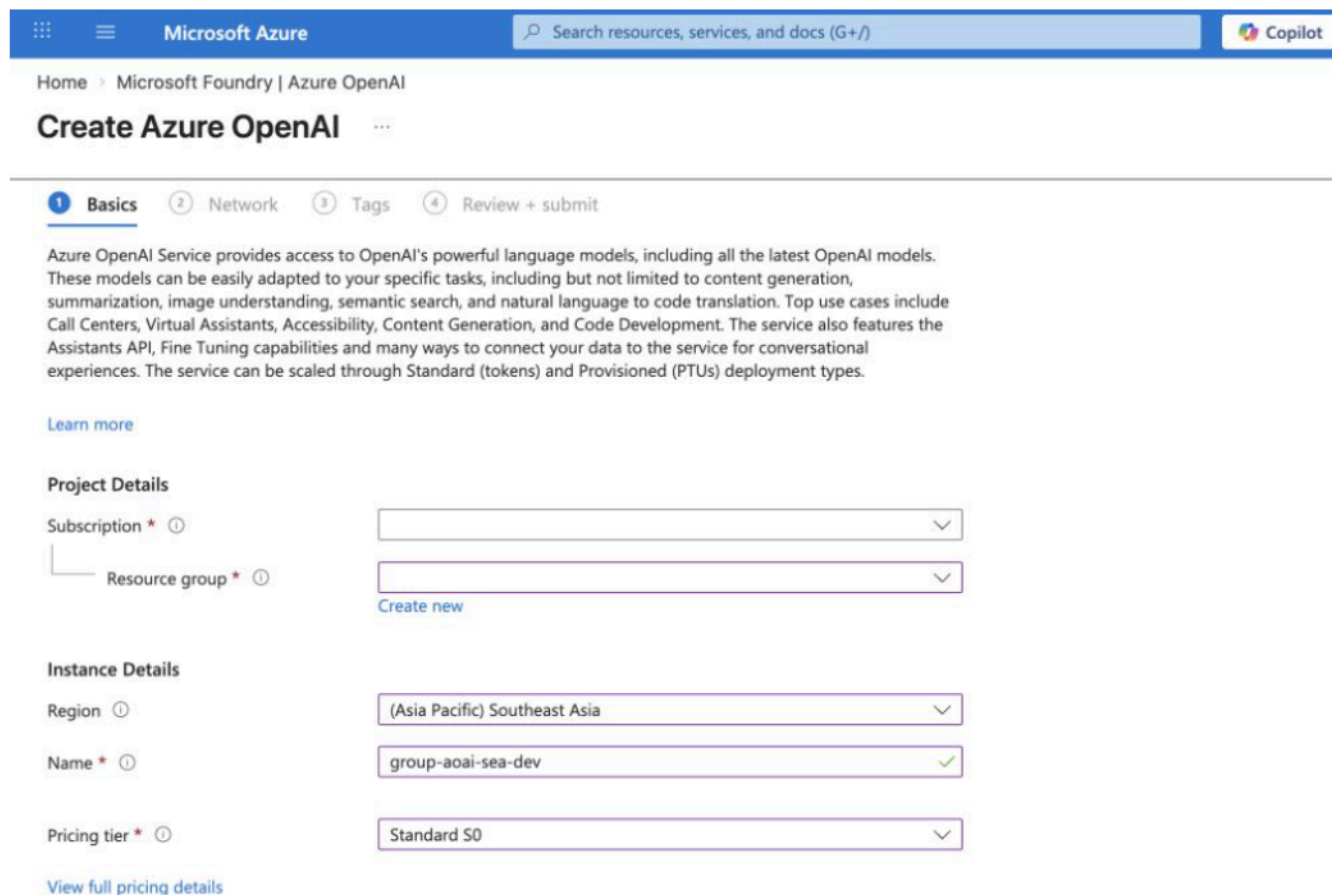
- It will open the Microsoft Foundry or Azure OpenAI page.
- To create instance, click **Create**.
- Then choose **Azure OpenAI**.



The screenshot shows the Microsoft Azure portal interface. At the top, there's a blue header with the Microsoft Azure logo, a search bar, and various icons. Below the header, the breadcrumb trail reads "Home > Microsoft Foundry". The main content area is titled "Microsoft Foundry | Azure OpenAI" with a sub-header "Bank Rakyat Indonesia". There are three buttons: "Filter resources by region", "Suggest services related to this list", and "List CLI commands to manage these resources". A search bar is present. Below the search bar, there's a list of services. The "Create" button is highlighted, and a dropdown menu is open, showing two options: "Foundry (Recommended)" and "Azure OpenAI". The "Azure OpenAI" option is highlighted with a red box. The "Foundry (Recommended)" option has a description: "Design, customize, and manage AI agents and models including OpenAI." The "Azure OpenAI" option has a description: "Use Azure OpenAI models to create, test, deploy your AI solutions end-to-end." The left sidebar shows a navigation menu with options like "Overview", "All resources", "Use with Foundry", "Foundry", "AI Hubs", "Azure OpenAI", "AI Search", "More services", "Classic AI services", and "Help".

Azure OpenAI: Create Instance

- Fill the project details in basics section.
- Choose your subscription, resource group, and region based on your company's policy.
- For the name, use lowercase letter with format: {group}-{instance}-{region}-{environment}.
For example: bsi-aoai-sea-dev
- Choose standard pricing tier
- Then, click **Next**.



The screenshot shows the 'Create Azure OpenAI' wizard in the Microsoft Azure portal. The breadcrumb trail is 'Home > Microsoft Foundry | Azure OpenAI'. The title is 'Create Azure OpenAI'. The wizard has four steps: 1. Basics (active), 2. Network, 3. Tags, and 4. Review + submit. A descriptive paragraph about the Azure OpenAI Service is provided, along with a 'Learn more' link. The 'Project Details' section includes dropdowns for 'Subscription' and 'Resource group', with a 'Create new' link next to the resource group dropdown. The 'Instance Details' section includes dropdowns for 'Region' (set to '(Asia Pacific) Southeast Asia'), 'Name' (set to 'group-aoai-sea-dev' with a green checkmark), and 'Pricing tier' (set to 'Standard S0'). A 'View full pricing details' link is at the bottom.

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home > Microsoft Foundry | Azure OpenAI

Create Azure OpenAI

1 Basics 2 Network 3 Tags 4 Review + submit

Azure OpenAI Service provides access to OpenAI's powerful language models, including all the latest OpenAI models. These models can be easily adapted to your specific tasks, including but not limited to content generation, summarization, image understanding, semantic search, and natural language to code translation. Top use cases include Call Centers, Virtual Assistants, Accessibility, Content Generation, and Code Development. The service also features the Assistants API, Fine Tuning capabilities and many ways to connect your data to the service for conversational experiences. The service can be scaled through Standard (tokens) and Provisioned (PTUs) deployment types.

[Learn more](#)

Project Details

Subscription * ⓘ

Resource group * ⓘ

[Create new](#)

Instance Details

Region ⓘ (Asia Pacific) Southeast Asia

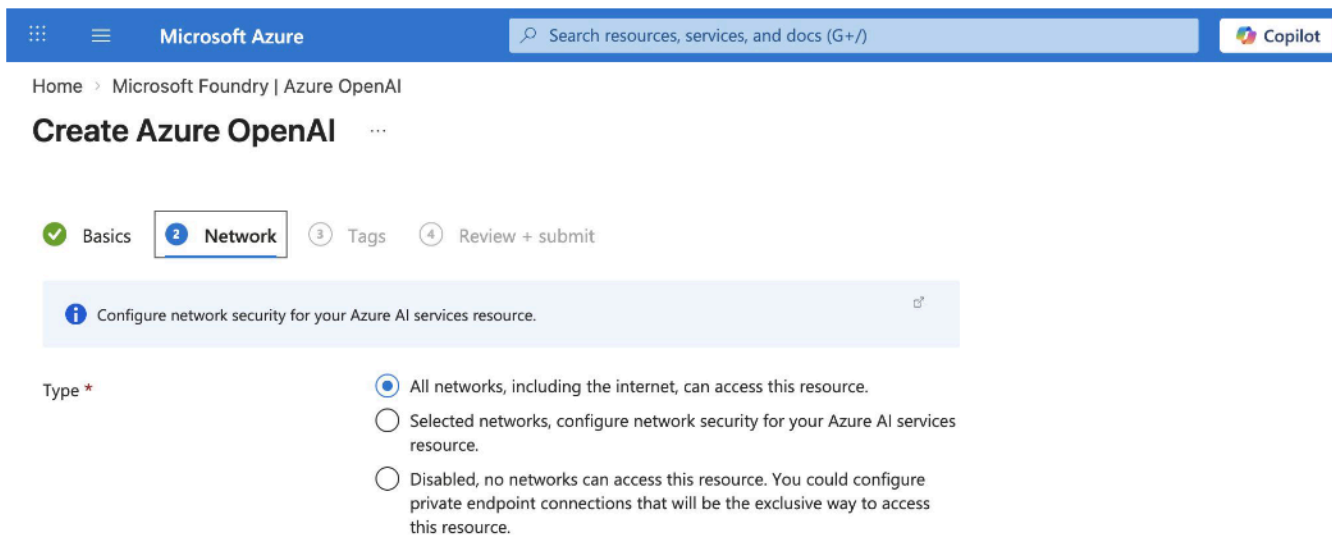
Name * ⓘ group-aoai-sea-dev ✓

Pricing tier * ⓘ Standard S0

[View full pricing details](#)

Azure OpenAI: Create Instance

- Configure the network security in network section.
- Choose all networks if there are no customize network security for your service.
- Then, click **Next**.



The screenshot shows the Microsoft Azure portal interface for creating an Azure OpenAI instance. The top navigation bar includes the Microsoft Azure logo, a search bar, and the Copilot icon. The breadcrumb trail indicates the path: Home > Microsoft Foundry | Azure OpenAI. The main heading is 'Create Azure OpenAI'. Below this, there are four steps in a sequence: 1. Basics (completed with a green checkmark), 2. Network (active, highlighted with a blue border and a blue circle with the number 2), 3. Tags, and 4. Review + submit. A light blue information box contains the text: 'Configure network security for your Azure AI services resource.' Below this, the 'Type' section has three radio button options: 'All networks, including the internet, can access this resource.' (which is selected), 'Selected networks, configure network security for your Azure AI services resource.', and 'Disabled, no networks can access this resource. You could configure private endpoint connections that will be the exclusive way to access this resource.'

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Home > Microsoft Foundry | Azure OpenAI

Create Azure OpenAI

1 Basics 2 Network 3 Tags 4 Review + submit

i Configure network security for your Azure AI services resource.

Type *

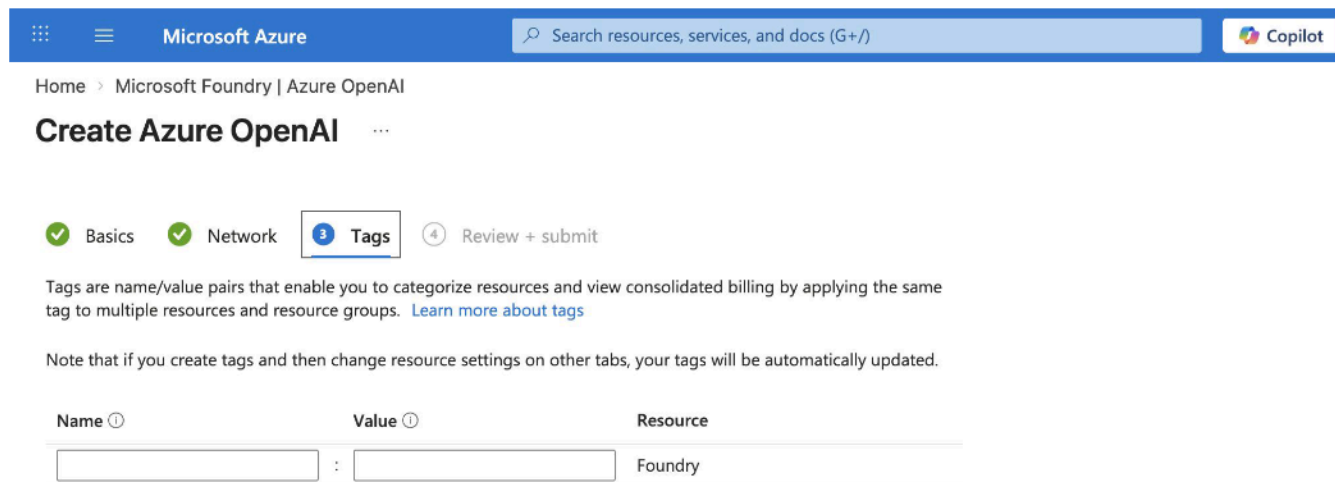
☒ All networks, including the internet, can access this resource.

☐ Selected networks, configure network security for your Azure AI services resource.

☐ Disabled, no networks can access this resource. You could configure private endpoint connections that will be the exclusive way to access this resource.

Azure OpenAI: Create Instance

- Configure the name/value in tags section.
- Tags are name/value pairs that enable you to categorize resource.
- Then, click **Next**.



The screenshot shows the 'Create Azure OpenAI' wizard in the Microsoft Azure portal. The 'Tags' step is selected, showing a table to define tags. The table has columns for Name, Value, and Resource. A single tag is pre-filled with 'Foundry' in the Resource column. The wizard progress shows steps 1 (Basics), 2 (Network), 3 (Tags), and 4 (Review + submit).

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Home > Microsoft Foundry | Azure OpenAI

Create Azure OpenAI

Basics Network **Tags** Review + submit

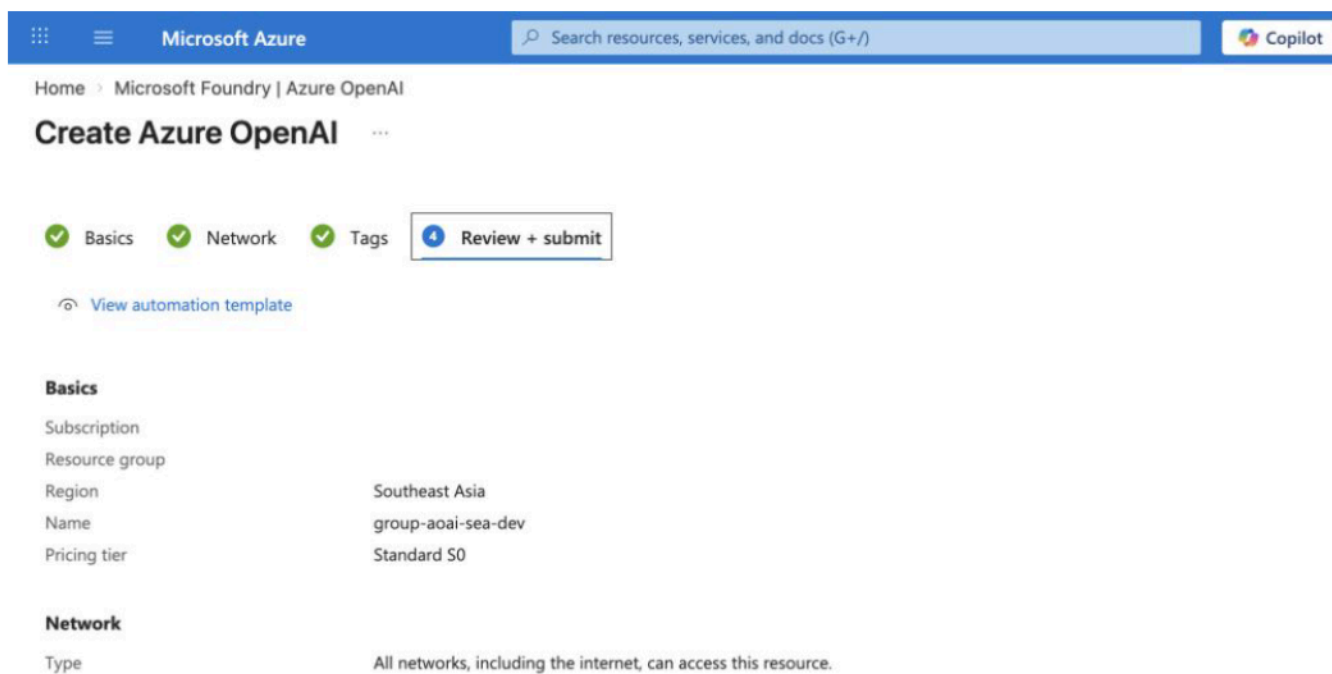
Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. [Learn more about tags](#)

Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.

Name	Value	Resource
		Foundry

Azure OpenAI: Create Instance

- Review the basics and network settings that have already been configured before submitting.
- If everything is correct, click **Submit**.



The screenshot shows the 'Create Azure OpenAI' page in the Microsoft Azure portal. The breadcrumb trail is 'Home > Microsoft Foundry | Azure OpenAI'. The page title is 'Create Azure OpenAI'. There are four steps: 'Basics' (checked), 'Network' (checked), 'Tags' (checked), and 'Review + submit' (active). A link 'View automation template' is present. The 'Basics' section shows: Subscription, Resource group, Region (Southeast Asia), Name (group-aoai-sea-dev), and Pricing tier (Standard S0). The 'Network' section shows: Type (All networks, including the internet, can access this resource.).

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home > Microsoft Foundry | Azure OpenAI

Create Azure OpenAI

Basics Network Tags Review + submit

[View automation template](#)

Basics

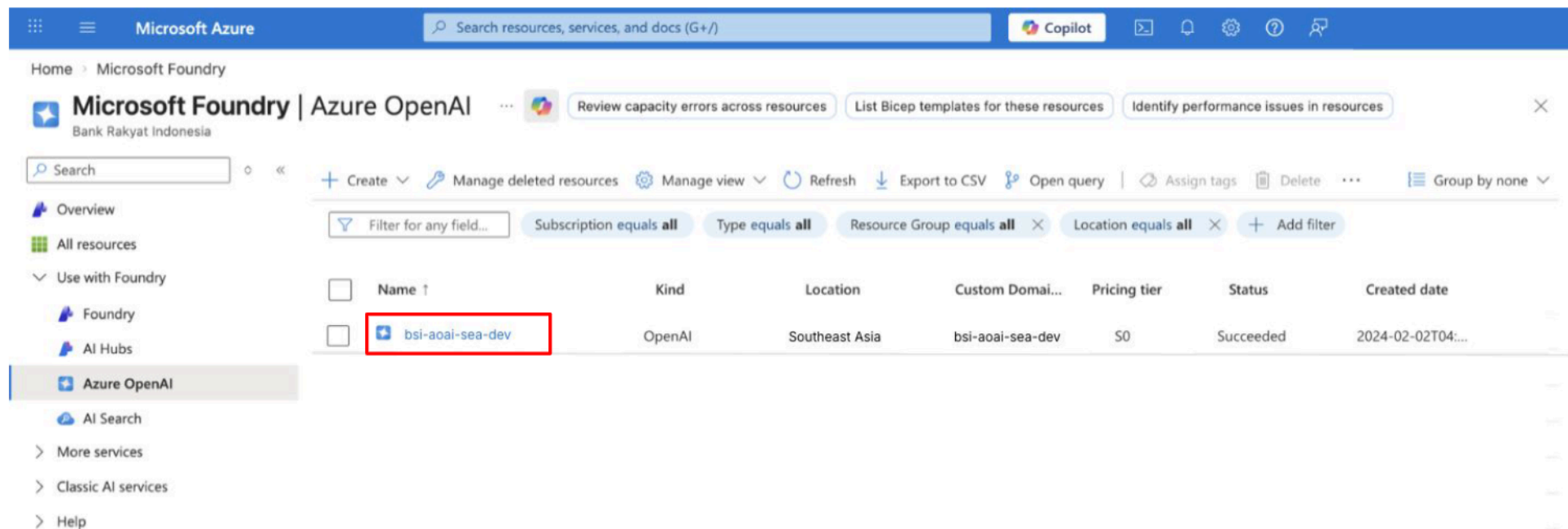
Subscription	
Resource group	
Region	Southeast Asia
Name	group-aoai-sea-dev
Pricing tier	Standard S0

Network

Type	All networks, including the internet, can access this resource.
------	---

Azure OpenAI: Get the API Key

- Open the Microsoft Foundry or Azure OpenAI page again.
- Then, click the instance that was created.

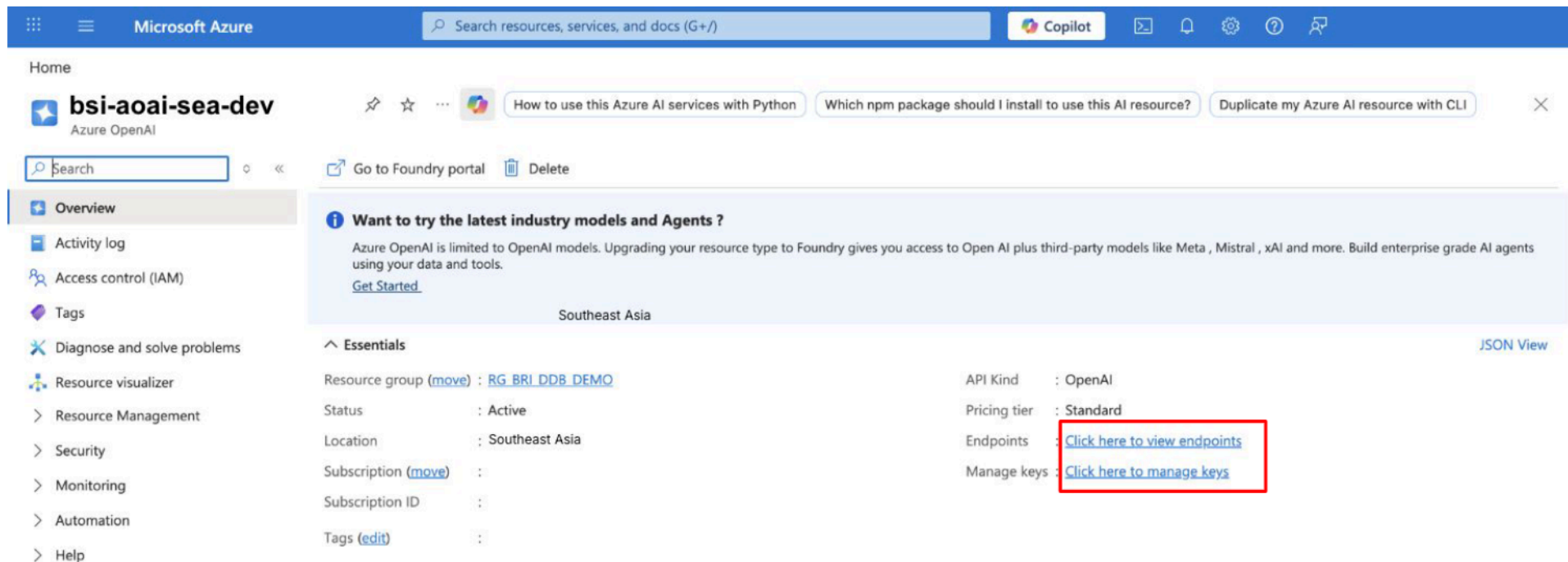


The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and the Copilot icon. The main content area displays the 'Microsoft Foundry | Azure OpenAI' page for 'Bank Rakyat Indonesia'. The left sidebar shows a navigation menu with options like Overview, All resources, Use with Foundry, Foundry, AI Hubs, Azure OpenAI (selected), AI Search, More services, Classic AI services, and Help. The main area features a table of resources. The first row in the table is highlighted with a red box, indicating the selected instance.

Name	Kind	Location	Custom Domai...	Pricing tier	Status	Created date
bsi-aoai-sea-dev	OpenAI	Southeast Asia	bsi-aoai-sea-dev	S0	Succeeded	2024-02-02T04:...

Azure OpenAI: Get the API Key

- It will open the Azure OpenAI Instance page.
- Then, click the **Endpoints** or **Manage keys**



The screenshot shows the Azure OpenAI instance page for 'bsi-aoai-sea-dev'. The page has a blue header with the Microsoft Azure logo and a search bar. Below the header, there's a navigation pane on the left with options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Resource Management, Security, Monitoring, Automation, and Help. The main content area displays the instance details for 'bsi-aoai-sea-dev' in the Southeast Asia region. A banner at the top of the main content area promotes upgrading to Foundry. Below the banner, there's a section titled 'Essentials' with details about the resource group, status, location, subscription, and tags. To the right of the Essentials section, there's a table of instance properties: API Kind (OpenAI), Pricing tier (Standard), Endpoints (Click here to view endpoints), and Manage keys (Click here to manage keys). The 'Endpoints' and 'Manage keys' links are highlighted with a red box.

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Home

bsi-aoai-sea-dev
Azure OpenAI

How to use this Azure AI services with Python

Which npm package should I install to use this AI resource?

Duplicate my Azure AI resource with CLI

Search

Go to Foundry portal

Delete

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Resource Management

Security

Monitoring

Automation

Help

Want to try the latest industry models and Agents ?

Azure OpenAI is limited to OpenAI models. Upgrading your resource type to Foundry gives you access to Open AI plus third-party models like Meta , Mistral , xAI and more. Build enterprise grade AI agents using your data and tools.

[Get Started](#)

Southeast Asia

Essentials

Resource group (move) : [RG BRI DDB DEMO](#)

Status : Active

Location : Southeast Asia

Subscription (move) :

Subscription ID :

Tags (edit) :

API Kind : OpenAI

Pricing tier : Standard

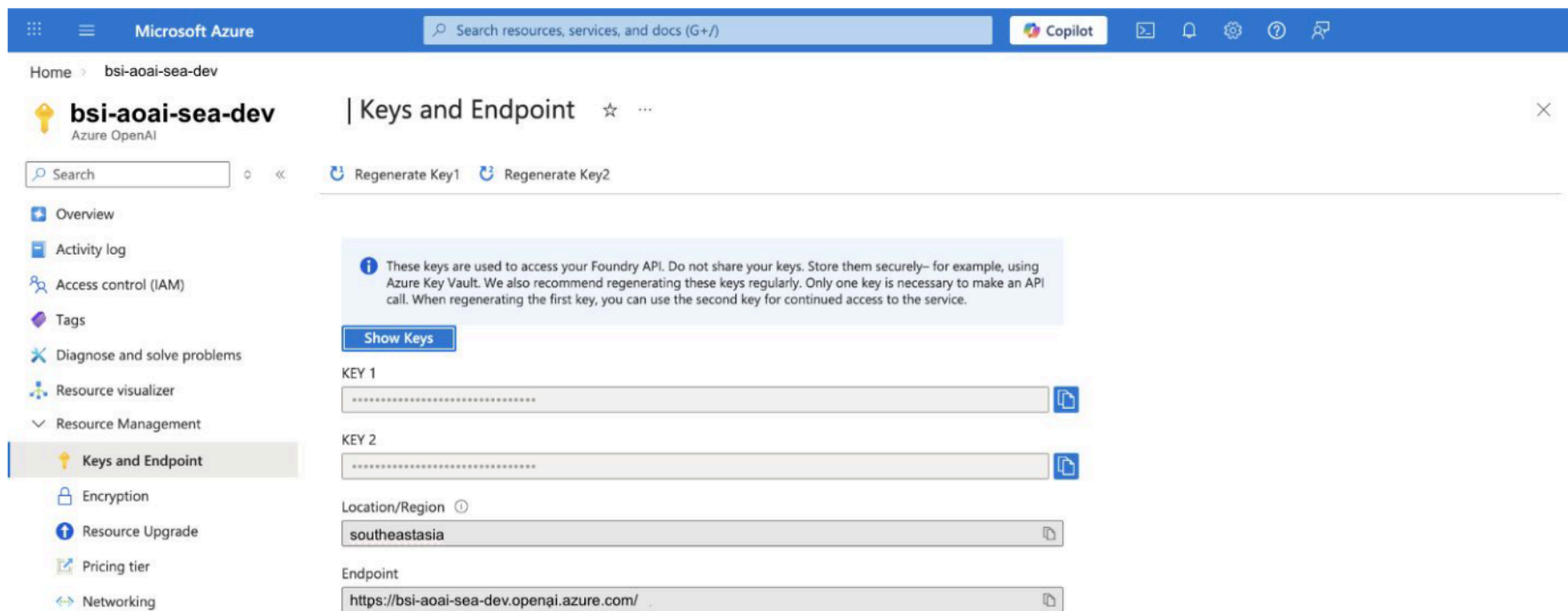
Endpoints : [Click here to view endpoints](#)

Manage keys : [Click here to manage keys](#)

JSON View

Azure OpenAI: Get the API Key

- It will open the keys and endpoint section.
- Click the copy icon in key 1 or key 2 for the **API Key**.
- Click the copy icon in endpoint for the **Endpoint**.



The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and the Copilot icon. The breadcrumb trail indicates the user is in the 'bsi-aoai-sea-dev' resource group. The main content area is titled 'Keys and Endpoint' and contains a search bar, 'Regenerate Key1', and 'Regenerate Key2' buttons. A left-hand navigation pane lists various resource management options, with 'Keys and Endpoint' selected. The main content area displays a warning message about key security, followed by a 'Show Keys' button. Below this, the 'KEY 1' and 'KEY 2' fields are shown with copy icons. The 'Location/Region' is set to 'southeastasia', and the 'Endpoint' is 'https://bsi-aoai-sea-dev.openai.azure.com/'.

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Home > bsi-aoai-sea-dev

bsi-aoai-sea-dev
Azure OpenAI

Keys and Endpoint

Search

Regenerate Key1 Regenerate Key2

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Resource Management

Keys and Endpoint

Encryption

Resource Upgrade

Pricing tier

Networking

These keys are used to access your Foundry API. Do not share your keys. Store them securely— for example, using Azure Key Vault. We also recommend regenerating these keys regularly. Only one key is necessary to make an API call. When regenerating the first key, you can use the second key for continued access to the service.

Show Keys

KEY 1

KEY 2

Location/Region ⓘ

southeastasia

Endpoint

https://bsi-aoai-sea-dev.openai.azure.com/

Embedding Generation



Embeddings: Overview

Meaning, not surface wording

An embedding is a numeric representation of text. Humans do not read the vector directly; systems use distance and similarity between vectors to compare meaning.

Text

Original sentence or document chunk

Embedding vector

Semantic representation for similarity math

Retrieve

Nearest matches

Why retrieval gets better

Semantic closeness lets the system find related content even when the query words do not literally match the source wording.

Why this matters architecturally

Embeddings connect the API layer to the retrieval layer. Without them, knowledge retrieval is often stuck with literal term overlap.

Embeddings: Model Selection

Default

text-embedding-3-small

Practical lower-cost default for many retrieval tasks.

Sample environment output uses 1536 dimensions.

Good when cost, storage footprint, and response speed matter more than squeezing for harder retrieval quality gains.

Selective

text-embedding-3-large

Larger representation for workloads where the extra cost is justified.

Sample environment output uses 3072 dimensions.

Useful when the retrieval task is demanding enough that the bigger vector footprint earns its keep.

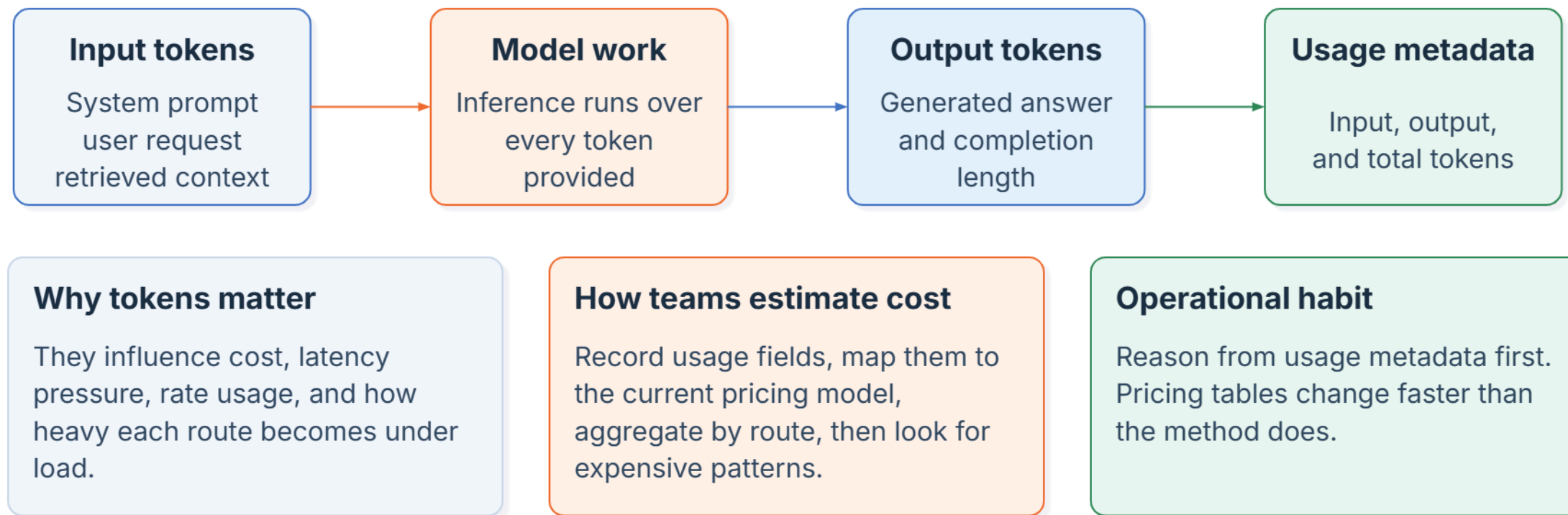
Selection rule

Choose based on workload quality needs, latency tolerance, storage footprint, and budget. Bigger is not automatically better.

Token Counting & Cost Estimation



Tokens and Cost Thinking



Azure OpenAI Model Pricing

Explore pricing options

Apply filters to customize pricing options to your needs.

Prices are estimates only and are not intended as actual price quotes. Actual pricing may vary depending on the type of agreement entered with Microsoft, date of purchase, and the currency exchange rate. Prices are calculated based on US dollars and converted using London closing spot rates that are captured in the two business days prior to the last business day of the previous month end. If the two business days prior to the end of the month fall on a bank holiday in major markets, the rate setting day is generally the day immediately preceding the two business days. This rate applies to all transactions during the upcoming month. Sign in to the [Azure pricing calculator](#) to see pricing based on your current program/offer with Microsoft. Contact an [Azure sales specialist](#) for more information on pricing or to request a price quote. See [frequently asked questions](#) about Azure pricing.

Region:	Currency:
East US	United States – Dollar (\$) USD

To check Azure OpenAI model pricing, visit the URL:
<https://azure.microsoft.com/en-us/pricing/details/azure-openai>

GPT-5.2

GPT-5.2 delivers the deep reasoning and expanded context handling necessary for building sophisticated AI agents capable of automating complex, long-running tasks across all business functions.

Model	Pricing (1M Tokens)
GPT-5.2 Codex Global	Input: \$1.75 Cached Input: \$0.18 Output: \$14
GPT-5.2 Global	Input: \$1.75 Cached Input: \$0.18 Output: \$14

Why Do LLMs Become Expensive

Oversized prompts

Irrelevant context raises input tokens without improving the answer.

Verbose outputs

Long completions cost more and can slow the route down.

Wrong model for the job

A more expensive model is wasteful when a smaller choice already works.

Repeated work

Recomputing embeddings or resending stable prefixes burns budget unnecessarily.

Practical response

Reduce prompt waste, right-size the model, reuse cached or stored artifacts, and watch route-level usage over time.

Processing Large Context Inputs



Large Context Risks and Mitigations

Risks to watch

More context can mean higher cost, slower requests, weaker signal-to-noise ratio, and harder reasoning about what the model actually needed.

Chunk

Break long text into units

Retrieve

Select only the relevant pieces

Cache

Reuse stable prompt prefixes

Generate

Answer with focused context

Mitigation principle

Give the model the best context, not the most context. Retrieval and chunking are architecture tools, not prompt-writing decoration.

Operational benefit

Focused inputs protect quality, reduce waste, and make large-context use cases easier to scale and monitor.

Prompt Caching in Practical Terms

What caching helps with

Prompt caching rewards repeated prefixes. Stable shared instructions or repeated prompt scaffolding can become cheaper and faster when the required conditions are met.

Request A

Stable prefix
+ user-specific
suffix



Request B

Same prefix
+ different suffix

Remember

The repeated beginning matters most.

Use it well

Only rely on caching when the prefix is truly stable.

Do not confuse

Caching is an optimization, not a substitute for prompt discipline.

Vector Database Workflow



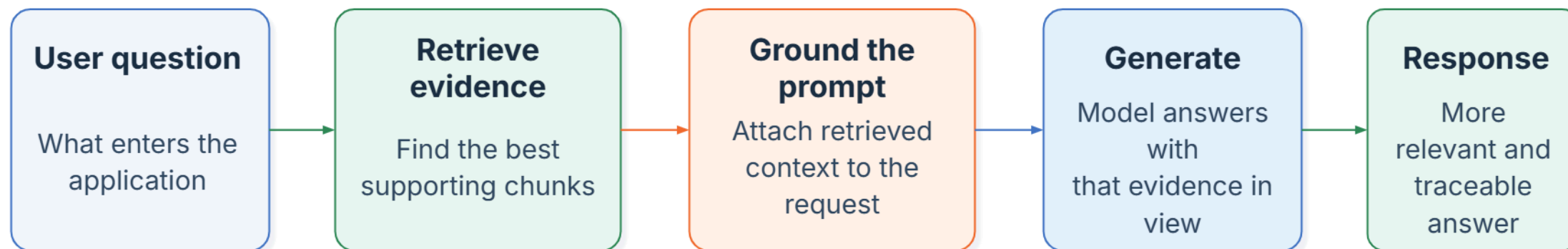
Vector Database Workflow



Why this sequence matters

RAG quality is shaped before generation starts. Weak chunking, stale content, or poor retrieval ranking can all lower answer quality even with a strong model.

RAG Pipeline



When direct generation is enough

If the prompt already carries the needed context and the task does not depend on changing external knowledge, you may not need retrieval.

When RAG earns complexity

If the answer must be grounded in private, fresh, or controlled documents, retrieval becomes part of the system design.

Why the flow stays teachable

RAG stays understandable when it is shown as retrieval, grounding, and then generation.

Direct Generation vs RAG

Simpler

Direct generation

- Use when general knowledge or provided input is sufficient
- Best for rewriting, summarizing, formatting, and ideation
- Simpler architecture (prompt → model)
- Faster and lower cost
- ⚠ May produce outdated or hallucinated answers

Prompt

Model

Grounded

Retrieval Augmented Generation (RAG)

- Use when answers require external, private, or up-to-date data
- Retrieves relevant context before generating responses
- Reduces hallucination and improves accuracy
- Supports controlled and auditable knowledge sources
- Output quality depends on retrieval + generation

Query

Retrieve context

Model

Retrieval Quality Checklist

- 1** Chunk content so each unit keeps one coherent idea.
- 2** Store useful metadata so retrieval can filter intelligently.
- 3** Keep indexed content fresh enough for the use case.
- 4** Use semantic retrieval when wording mismatch is common.
- 5** Combine exact matching and vector similarity when identifiers matter.
- 6** Inspect retrieved evidence before blaming the model for weak answers.

Guided Hands-on



Token Optimization

Learning Goals

By the end of this session, participants will be able to:

- Analyze how token usage affects LLM cost and performance.
- Optimize prompts using token estimation and reduction techniques.
- Handle long text using chunking and retrieval strategies.
- Build more efficient and scalable LLM workflows

Setup

Participants can use:

- Google Colaboratory:
<https://colab.research.google.com/drive/15aDaAVcBAO6FVsYB3PpraT4L7u4JJZsl?usp=sharing>

Assignment



Token Optimization with Azure OpenAI

Objective

Apply token optimization techniques using real LLM API usage and observe actual token consumption.

Scenario

You are building a Q&A assistant for company documents using Azure OpenAI.

Your challenge:

Reduce token usage while still getting a good answer from the model.

Requirements

Participants must:

- Use Azure OpenAI API Key
- Run code in Google Colab
- Use the provided helper functions (`estimate_tokens`, `chunk_text`, `simple_retrieve`)

Step 1: Setup Azure OpenAI

```
!pip install openai
```

```
from openai import OpenAI

client = OpenAI(
    api_key="YOUR_API_KEY",
    base_url="YOUR_AZURE_ENDPOINT/openai/v1/"
)

MODEL_NAME = "YOUR_CHAT_DEPLOYMENT"
```


Step 2: Baseline (Inefficient Way)

```
all_text = " ".join([doc.content for doc in KNOWLEDGE_BASE])

response = client.responses.create(
    model=MODEL_NAME,
    input=f"Answer this question based on the context:\n{all_text}\n\nQuestion: How to handle large prompts?"
)

print(response.output_text)
print(response.usage)
```

Task:

- Combine ALL documents
- Send to LLM
- Record token usage

👉 Save:

- input_tokens
- output_tokens
- total_tokens

Step 3: Optimized Approach

```
query = "How to handle large prompts?"

retrieved_docs = simple_retrieve(query, KNOWLEDGE_BASE, top_k=1)
retrieved_text = " ".join([doc.content for doc in retrieved_docs])

response_opt = client.responses.create(
    model=MODEL_NAME,
    input=f"Answer this question based on the context:\n{retrieved_text}\n\nQuestion:
{query}"
)

print(response_opt.output_text)
print(response_opt.usage)
```

Task:

- Use `simple_retrieve()` → top 1-2 docs only
- Send only selected content

👉 Save:

- `input_tokens`
- `output_tokens`
- `total_tokens`

Step 4: Compare Results

Answer this:

1. How many tokens were used in:
 - Baseline
 - Optimized
2. How much token reduction did you achieve?
3. Compare:
 - Response quality
 - Response speed (feel-based)

Approach	Input Tokens	Output Tokens	Total Tokens
Baseline	xxx	xxx	xxx
Optimized	xxx	xxx	xxx



Thank You